From Subscription Prediction to Smart Marketing Targeting

Complete Project Synthesis | Portuguese Bank Marketing | Group 1

This document independently synthesizes the full team project. Individual experiments are described as team work, not represented as the author's personal experiments.

Executive summary

The objective was to prioritize customers likely to subscribe to a term deposit before a marketing call. The team analyzed 45,211 historical contacts, compared four classification algorithms, tested feature engineering and imbalance remedies, and selected a leakage-aware XGBoost ranking model.

Under five-fold shuffled stratified cross-validation, the final strict pre-call XGBoost model achieved MCC 0.4211 ± 0.0219. Retrospective out-of-fold ranking placed 62.19% of observed subscribers within the top-ranked 20% of customers. Forward-time tests showed substantial performance deterioration, making prospective validation essential.

1. Business question and decision context

The bank has a limited contact-center budget. Rather than call all customers indiscriminately, the intended decision is to rank potential contacts using information available at the time of selection. Prediction is not causation: a high predicted score does not establish that a call will create a subscription.

Success should ultimately be measured through a future pilot with cost per contact, incremental conversion, capacity constraints, and customer experience metrics; the available dataset does not support a verified ROI estimate.

2. Dataset, audit and target

Source: Portuguese Bank Marketing bank-full.csv, containing 45,211 records, 16 original predictors, and binary subscription outcome y. There were 5,289 yes cases (11.7%) and 39,922 no cases (88.3%). No nulls or duplicate records were reported.

Literal unknown categories were retained rather than treated as null. The special pdays value −1 indicates no prior contact in an earlier campaign. The class imbalance makes raw accuracy an unreliable primary performance measure.

3. Exploratory analysis: descriptive patterns

Previous campaign success was associated with a 64.73% observed subscription rate versus approximately 9.16% for no prior contact. Students had approximately 28.7% and retired customers approximately 22.8% subscription rates.

Subscription was 14.6% at one contact versus 3.93% for 11+ contacts. Call duration showed a particularly strong association: under one minute, 0.19% subscribed; over ten minutes, 48.32% subscribed. These are historical associations and may reflect selection, campaign phase, and outcome-dependent processes.

4. Feature availability and leakage

Call duration is recorded after a conversation begins and cannot be used to select customers before calling. The initial common algorithm comparison removed duration but retained campaign, the count of contacts during the current campaign. The strict final pre-call feature set removed both duration and campaign.

This distinction explains why some earlier experiments are labelled without-duration rather than strict pre-call. The dataset also contains date- and campaign-phase proxies, particularly month, day and contact channel, which can make random-fold validation optimistic for future deployment.

5. Experimental methodology

The initial four-algorithm comparison used original predictors without duration, no feature engineering, no class imbalance treatment, threshold 0.5, and five stratified shuffled folds with random_state 42. Preprocessing, including categorical one-hot encoding, was performed within training folds.

Matthews correlation coefficient (MCC) was the primary model selection metric because it summarizes true and false positives and negatives. Majority-class prediction gives MCC 0 even though its accuracy is approximately 88.3%. The feature-engineering experiments and imbalance experiments used their own stated model settings; numerical scores across those stages should not be interpreted as perfectly matched head-to-head tests.

6. Algorithm comparison

Algorithm	Without duration: mean MCC	With duration: diagnostic MCC
Logistic Regression	0.3021 ± 0.0194	0.4267 ± 0.0110
Decision Tree, tuned	0.3166 ± 0.0223	0.4475
Random Forest, tuned	0.3305	0.4699
XGBoost, tuned	0.355 ± 0.014	0.515 ± 0.016

The RF standard deviation differs across the team's report sections; the common-comparison table therefore states its verified mean without asserting a single reconciled SD.

7. Decision Tree: controlling overfitting

An untuned tree had without-duration MCC near 0.215. Hyperparameter tuning improved this to approximately 0.317. Average leaves declined from 4,763 to 40, a 99.2% reduction, and training MCC fell from 1.000 to approximately 0.338. The model became substantially simpler and less overfit, although recall remained around 20.6%.

8. Random Forest: aggregation and limited tuning gain

Tuned Random Forest produced without-duration MCC 0.3305, only slightly above the reported untreated baseline 0.3302. With duration, tuned MCC was approximately 0.4699. This stage suggested that ensemble aggregation improved performance over a single tree, but further hyperparameter tuning alone contributed little under the specified untreated comparison.

9. XGBoost: selected model family

Tuned XGBoost led the common no-duration comparison with MCC 0.355, compared with 0.331 for Random Forest, 0.317 for Decision Tree, and 0.302 for Logistic Regression. The tuned settings included 624 trees, learning rate 0.025, maximum depth 7, min_child_weight 12, subsample 0.66, colsample_bytree 0.59, gamma 0.92, and L2 regularization 2.56.

The choice was to continue optimizing XGBoost, not to claim that XGBoost would necessarily outperform every other algorithm after all possible algorithm-specific feature and imbalance adjustments.

10. Feature engineering and selection

The team evaluated age bands, balance transformations, contact and previous-campaign history, interaction features and selection methods. In a separate fixed-setting XGBoost design (five folds × three seeds), the strict raw feature set achieved MCC 0.341; strict engineered features reached 0.346, and interaction-plus-selection variants around 0.348.

With class weight 4, strict raw and strict engineered variants were essentially tied (0.4225 versus 0.4220). A separate nested tuning check showed a small difference (0.3505 versus 0.3522). These marginal and inconsistent gains supported keeping the simpler strict raw predictors.

The correct bank-full dataset does not contain bank-additional-full variables such as euribor3m or emp.var.rate; these are intentionally excluded from this synthesis.

11. Class imbalance experiments

The team compared tuned XGBoost without duration under positive-class weights 2, 3, 4, and 7.55, plus oversampling ratios 0.5 and 1.0. Any resampling was restricted to training folds.

Method	Mean MCC	Recall
Untreated	0.3525	0.2462
Weight 2	0.3964	—

Weight 3	0.4193	—
Weight 4	0.4230	0.5224
Weight 7.55	0.3805	—
Oversampling 0.5	0.4218	0.5060
Oversampling 1.0	0.3839	—

Weight 4 improved recall from 24.6% to 52.2% while increasing MCC. Oversampling at ratio 0.5 performed similarly; the 0.0012 MCC difference is too small to present as a decisive superiority. Stronger weighting or full oversampling reduced MCC.

12. Final model specification and results

Final configuration: tuned XGBoost, positive-class weight 4, 14 original strictly pre-call predictors (excluding duration and campaign), five-fold stratified shuffled CV with seed 42, and threshold 0.5.

Final fold MCC values were 0.3974, 0.4558, 0.4104, 0.4162, and 0.4256. Mean MCC was 0.4211 ± 0.0219; precision 0.4693, recall 0.5158, F1 0.4914. This is slightly below the 0.4230 from the separate imbalance experiment, while meeting the stricter input availability requirement.

13. Historical targeting value

Out-of-fold (OOF) scores were generated for every record by a model fitted on the other folds. Ranked OOF scores enable retrospective lift and gains analysis. Scores from a class-weighted model should not be assumed calibrated probabilities.

Top share	Customers	Observed yes	Share of all yes	Lift
10%	4,522	2,319	43.85%	4.38×
20%	9,043	3,289	62.19%	3.11×
30%	13,564	3,729	70.50%	2.35×
40%	18,085	4,085	77.24%	1.93×
50%	22,606	4,394	83.08%	1.66×

Interpretation: the historical top-ranked 20% contained 62.19% of observed subscribers. This is not a verified claim that contacting 20% in a future campaign will generate or capture that share.

14. Interpretation with SHAP

Model explanations emphasized previous campaign outcome, month, contact channel, job, day, balance, and housing status. Historical previous-campaign success had a positive SHAP effect around +1.39 in log-odds units (approximately fourfold odds), conditional on the fitted model.

Month, contact and day together contributed roughly 49% of mean absolute SHAP magnitude. They may represent campaign timing and historical data collection patterns rather than stable levers that a bank can change. SHAP quantifies model attribution, not causal effect.

15. The most important limitation: temporal drift

The historical data are ordered in time and exhibit strong drift: observed subscription rates were about 5.1% in 2008, 17.1% in 2009, and 51.6% in 2010. Shuffled cross-validation mixed campaign periods and yielded ROC AUC around 0.80, whereas forward-block evaluation was around 0.57.

Historical top-decile lift of 4.38× in shuffled OOF testing fell to roughly 1.3–1.5× in forward testing. Accordingly, the final shuffled-CV performance is useful as an internal benchmark, but should not be presented as a reliable prospective deployment forecast.

16. Recommendations and operational blueprint

First, repeat evaluation on a genuinely later campaign using only features known at the intended scoring time. Second, choose a calling cutoff based on staffing capacity, contact cost, customer constraints and expected value rather than assuming 0.5 is the optimal operational threshold.

Third, pilot the ranked list against a randomized or carefully designed control group. Track incremental conversion, cost per incremental subscription, contact rate, fairness across customer segments, and drift in input variables. Fourth, retrain and recalibrate only after collecting reliable new campaign outcomes.

17. What the project establishes — and does not

Established: a complete historical data audit, exploratory analysis, controlled four-model comparison, feature availability analysis, feature engineering tests, class imbalance studies, interpretable final XGBoost ranking, and strong evidence of temporal validation risk.

Not established: causal effects of customer characteristics or call timing, a validated financial ROI, calibrated subscription probabilities, or reliable performance on future unseen campaigns. These require additional evidence and prospective testing.

18. Suggested examiner Q&A

Why MCC? It is sensitive to all confusion-matrix cells and is more meaningful than accuracy for an 11.7% positive class.

Why remove duration? It is unavailable when deciding whom to call. Why remove campaign? The recorded total may include current-call activity and therefore fails the strict pre-call availability contract.

Why XGBoost? It led the matched untreated algorithm comparison; later experiments optimized that chosen family.

Why no engineered features in the final model? Marginal improvements were not consistent, and the simpler raw set performed comparably under class weighting.

Why not deploy immediately? Forward-time validation dropped from ~0.80 to ~0.57 AUC; prospective stability and economic value remain unproven.

Source and authorship note

Based on the team's uploaded final project report and 14-slide presentation, including reported model experiments and charts. The source material contains inconsistent standard deviations for some baseline models and an incorrect pasted feature-engineering narrative referring to a different bank dataset; this synthesis avoids those unsupported variables and does not invent reconciled values.

This is an original consolidation of team findings. It should not be used to attribute all experiments to a single individual.